LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	The procedure of Gear cutting in Milling machine
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	The procedure of Gear cutting in Milling machine
5. Specific Objectives	31.1.1.1 Trainee will be able to perform various general engineering works

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of gear cutting setup, safety checks, and cutter selection; Instructor supervision during setup verification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a milling machine?
(b) Name two types of gear cutters used in milling.
(c) What safety precaution is essential before starting a milling machine?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Selection of appropriate gear cutter • Workpiece mounting and alignment • Machine setup and parameter selection • Cutting speed and feed rate calculation • Initial roughing cuts execution	L & Ds	PPT, IB	15 min
(b)	• Finishing cuts and dimensional accuracy • Deburring and surface inspection • Measurement using vernier calipers • Identification of common cutting errors • Proper tool handling and storage	L & Ds	PPT, PDF	20 min
(c)	• Final quality assessment criteria • Documentation of work performed • Machine shutdown and cleaning procedures • Review of safety protocols observed • Preparation for next operation sequence	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the importance of correct cutter selection in gear cutting.
(b) Describe the steps taken to ensure dimensional accuracy during finishing cuts.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between gear cutting in a milling machine versus a shaping machine, referencing LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various types of common hand-held tools, power tools and their correct usage	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min